

# LOOPS

A structured, question-by-question collection of Python loop exercises

75

WHILE LOOPS

75

WHILE TRUE

75

FOR LOOPS

**Author:** Mirza Muhammad Ariz Baig

**Language:** Python 3

**Repository:** [github.com/mirzamuhammarizbaig/LOOPS](https://github.com/mirzamuhammarizbaig/LOOPS)

**Total exercises:** 225 standalone scripts + companion notes PDF

# About This Repository

**LOOPS** is a personal practice bank built while learning Python's iteration and control-flow constructs. Instead of one long notebook, every question lives in its own `.py` file, which makes it easy to run, tweak, and re-use individual exercises without scrolling through unrelated code.

The repo is organized around three core loop patterns — `for`, `while`, and `while True` — moving from simple counters and validation loops up to small interactive programs like an ATM simulator and menu-driven console apps.

💡 This repo is meant as a learning log and quick-reference bank — grab any script, read it, run it, and try to rewrite it from scratch as practice.

## Folder Structure

```
LOOPS/
├── for Loops/                                75 scripts (Q1 – Q75)
│   ├── Q1.py      ← Q1-Q19 coming soon
│   ├── Q20.py
│   └── ...
├── While Loops/                                75 scripts (Q1 – Q75)
│   ├── Q1.py
│   ├── Q2.py
│   └── ...
├── While True/                                75 scripts (Q1 – Q75)
│   ├── Q1.py
│   ├── Q2.py
│   └── ...
└── Python notes loops and conditional statement.pdf
    Handwritten / typed notes companion
```

## What's Inside Each Section

| Folder                    | Focus                               | Examples of what's covered                                                                                      |
|---------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>for Loops/</code>   | Counting & iteration                | Range-based counting, input validation loops, accumulators (e.g. counting positive numbers entered by the user) |
| <code>While Loops/</code> | Condition-driven repetition         | Menu-driven programs, an ATM balance/withdraw/deposit simulator, running totals                                 |
| <code>While True/</code>  | Infinite loops with explicit breaks | Input-until-exit patterns, sentinel-controlled loops, small interactive prompts                                 |

### 🔗 Sample — `While Loops/Q1.py`

```
# A simple ATM-style menu built with a while loop
balance = 1000
running = True

while running:
    choice = input("1.Check Balance\n2.Withdraw\n3.Deposit\n4.Exit\nChoose an Option: ")

    if choice == "1":
        print(f"Your account balance is ${balance}")
    elif choice == "4":
        print("Thank you for using our ATM")
        running = False
```

## Getting Started

Clone the repo and run any script directly with Python 3 — no external dependencies required.

```
git clone https://github.com/mirzamuhammadarizbaig/LOOPS.git
cd LOOPS
```

[Python 3](#)[No dependencies](#)[Standalone scripts](#)[Beginner friendly](#)

## Why This Repo Exists

---

- Build muscle memory for loop syntax and control flow through repetition
- Keep every exercise isolated and runnable on its own
- Create a searchable reference to revisit specific patterns later
- Track progress as part of a self-taught path into programming

## Possible Next Steps

---

- Add short problem statements as comments at the top of each file
- Group related questions with tags (validation, simulation, math, strings)
- Add a few nested-loop and list/dictionary-based exercises